

AI-Based Geotechnical Report Review & Foundation Recommendation Platform

Purpose

Develop an AI-enabled platform that automatically reviews geotechnical investigation reports, extracts engineering parameters, performs code-based calculations, and recommends suitable foundation solutions for wind turbine and other infrastructure projects.

1. Objectives

- Automate interpretation of soil investigation reports.
- Extract geotechnical parameters from PDF/DOC reports.
- Calculate Safe Bearing Capacity (SBC), immediate and long-term settlements.
- Evaluate dynamic soil properties and rotational stiffness.
- Recommend suitable foundation type and governing design criteria.
- Generate engineering summary reports.

2. Functional Scope

- OCR and AI extraction of borehole logs, lab data and groundwater.
- Validation of extracted values.
- Engineering calculations as per IS 6403, IS 1904, IS 8009, IS 5249 and configurable DNV/IEC requirements.
- Foundation suitability logic based on SBC, settlement, and dynamic criteria.
- Export results to Word/PDF/Excel.

3. AI Workflow

- Input soil report.
- AI parameter extraction.
- Engineering calculations.
- SBC and settlement verification.
- Foundation suitability assessment.
- Final recommendation.

4. Required Inputs

- Soil report (PDF/DOC).
- Foundation dimensions.
- Loads and moments.
- Groundwater level.
- Project location.
- Applicable design code.

5. Outputs

- Extracted soil profile.
- SBC (shear and settlement governed).
- Immediate and consolidation settlement.
- Dynamic soil properties.
- Rotational stiffness.
- Recommended foundation option (20 SBC/40 SBC or custom).
- Ground improvement recommendation.
- One-page engineering summary.

6. Vendor Deliverables

6.1 AI-Based Report Processing

- Upload soil investigation reports (PDF, Word, scanned documents).
- OCR and AI-based extraction of geotechnical parameters.
- Automatic engineering calculations based on configured design standards.
- Automated foundation recommendation with governing design criteria.

6.2 Engineering Calculation & Recommendation Engineer

- Automatic Safe Bearing Capacity (SBC) calculations based on shear and settlement criteria.
- Immediate and long-term settlement calculations.
- Dynamic soil property calculations.
- Rotational stiffness calculations.
- Automatic foundation recommendation (20 SBC, 40 SBC, or other configured foundation types).
- Identification of the governing design criterion.

6.3 Reengineering Recommendation Module

If the recommended SBC at the founding level does not satisfy the project requirement, the system shall:

- Automatically evaluate deeper soil strata.
- Recalculate SBC and settlement at successive founding depths.
- Recommend the minimum reengineering depth at which the required SBC is achieved.
- Recommend the excavation/removal depth for unsuitable soil.
- Indicate the replacement depth required for engineered fill, if applicable.
- Generate a reengineering summary report with the engineering basis for the recommendation.

6.4 Engineering Approval Workflow

The system shall include a configurable approval workflow.

- AI generates the engineering recommendation.
- The recommendation shall remain in **Pending Approval** status.
- The designated reviewer/approver shall review:
 - Extracted soil parameters
 - SBC calculations
 - Settlement calculations
 - Foundation recommendation
- The reviewer shall have the option to:
 - Approve
 - Reject
 - Request modification
- Only approved recommendations shall be considered final and available for downstream users.
- The system shall maintain complete approval history with reviewer name, comments, date, and revision number.

6.5 Soil Report Repository

The application shall maintain a centralized repository of all uploaded reports.

Features shall include:

- Customer-wise folder structure.
- Project-wise organization.
- Search by:
 - Customer

- Project
 - Location ID
 - Borehole ID
 - Date
 - Report number
 - Version control for revised reports.
 - Storage of original uploaded report.
 - Storage of AI-generated recommendation report.
 - Storage of final approved engineering report.
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6.6 Consolidated Engineering Register

The application shall automatically generate a consolidated register for each customer.

The register shall be exportable to Excel and include:

- Customer Name
- Project Name
- Location ID
- Borehole Number
- Foundation Level
- Recommended SBC
- Governing Design Criteria
- Settlement
- Recommended Foundation Type
- Reengineering Required (Yes/No)
- Recommended Reengineering Depth
- Approval Status
- Reviewer Name
- Approval Date

The register shall automatically update whenever a report is approved.

6.7 User Management

- Role-based access control.
 - Administrator.
 - Engineering Reviewer.
 - Engineering Approver.
 - Site Civil Team (Read Only).
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6.8 Dashboard

Interactive dashboard showing:

- Reports Uploaded
- Pending Review
- Pending Approval
- Approved Reports
- Rejected Reports
- Reports Requiring Reengineering
- Customer-wise project statistics
- Monthly report trends

6.9 Audit Trail

The application shall maintain a complete audit trail of:

- Upload history
- AI extraction changes
- Manual edits
- Engineering calculations
- Approval history
- Report downloads
- User activity

6.10 Deliverables

- Web-based application
- AI document processing engine
- Engineering calculation engine
- OCR module
- Approval workflow
- Customer repository
- Consolidated Excel generation
- Admin dashboard
- API integration
- User manual
- Source code
- Installation guide
- Testing and deployment support

7. Non-functional Requirements

- 95%+ extraction accuracy.
- Processing time <1 minutes/report.
- Secure storage.

8. Future Enhancements

- Learning from reviewer corrections.
- Multi-country code support.
- LLM chatbot for report queries.

9. Expected Benefits

- Reduce review time by >80%.
- Improve consistency.
- Reduce manual errors.
- Standardize engineering decisions.
- Create searchable geotechnical database.